PRIOR TO 1920, THESE ALL MEANT THE SAME THING:

THE MILKY WAY = THE GALAXY = THE UNIVERSE

1920: SHAPLEY – CURTIS DEBATE

SPIRAL NEBULA "ISLAND UNIVERSES" (GALAXIES)

1923: EDWIN HUBBLE (MT. WILSON OBSERVATORY)

TAKES SEVERAL IMAGES OF M31.

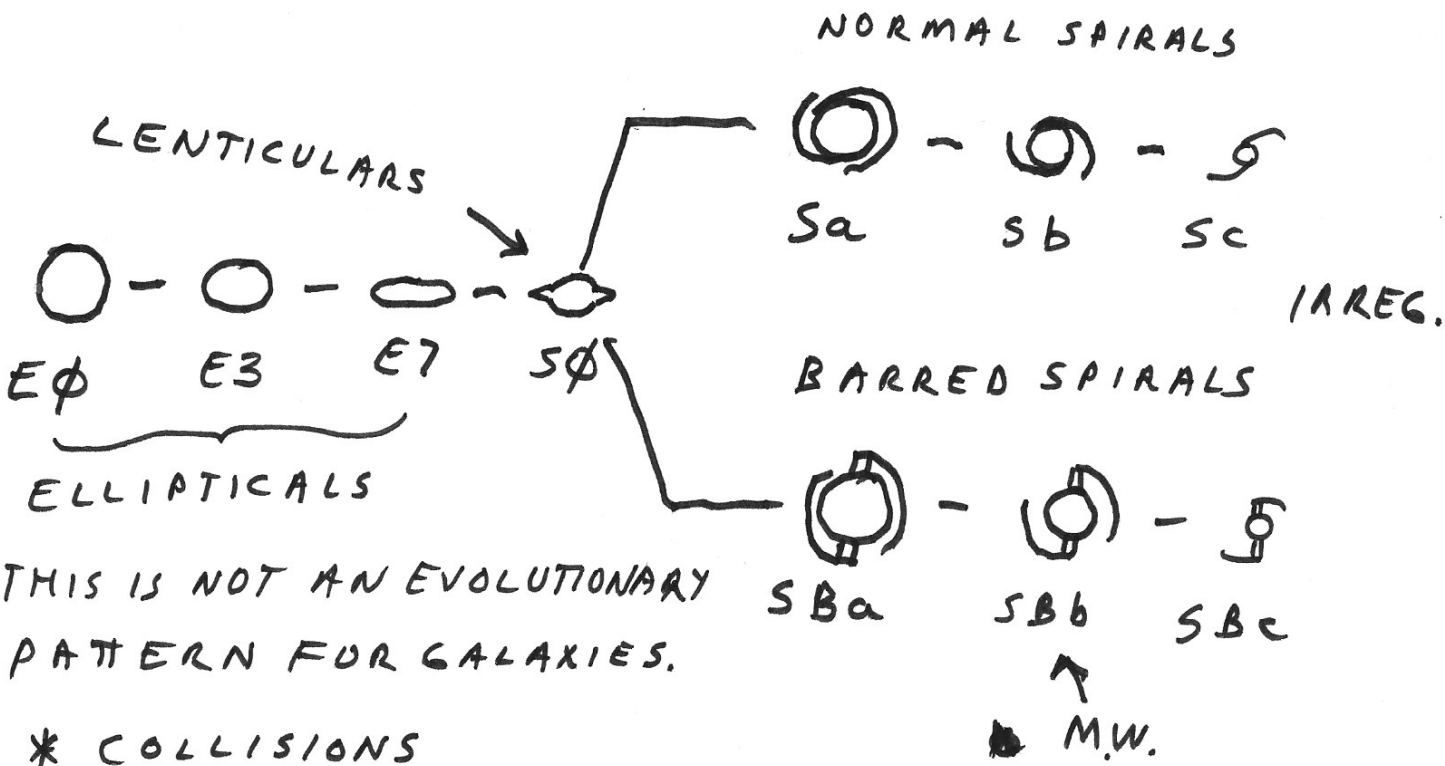
FOUND SEVERAL CEPHEIDS IN IT ☺

GETS A DISTANCE OF OVER 1 MILLION ly

(TODAY: 2.5 MILLION YEARS → DUST).

M31: ANDROMEDA GALAXY

TUNING FORK DIAGRAM (CLASSIFY GALAXIES)



THIS IS NOT AN EVOLUTIONARY PATTERN FOR GALAXIES.

* COLLISIONS

ELLIPTICALS (MOST)

CAN BE BOTH THE SMALLEST GALAXIES (DWARFS - 10^7 STARS) UP TO THE LARGEST GALAXIES (GIANTS: WELL OVER 10^{12} STARS).

SPIRALS

TEND TO BE MEDIUM-SIZED GALAXIES (A FEW 10^{11} STARS).

NO NEBULAE,
NO STAR FORMATION,
ONLY OLD, LOW MASS,
G, K, M, POP II STARS.

N.B. & HALO

DISK: NEBULAE,
STAR FORMATION,
YOUNG, HIGH MASS,
O, B, POP. I STARS.

BEST FOR LIFE:

IRREGULARS:
TYPICALLY VERY SMALL, NEAR MAJOR GALAXIES, TIDAL FORCES CAUSE THE IRREGULAR SHAPE AND ALSO TRIGGER LOTS OF STAR FORMATION (BLUE).

DURING THE 1920S, HUBBLE GETS, FOR MANY GALAXIES

1. DISTANCE (CEPHEIDS → P-L RELATIONSHIP). d (Mpc)
2. RADIAL VELOCITY (DOPPLER SHIFT). v (km/s)

- FOR THE ALMOST 50 "NEARBY" GALAXIES: R & B SHIFTS.
- FOR THE "DISTANT" GALAXIES: R SHIFTS ONLY.

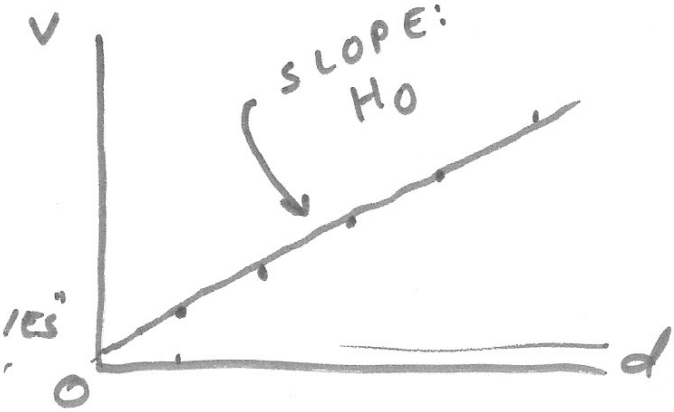
IGNORE THE NEARBY GALAXIES. FOR ALL OTHER GALAXIES: v → RECESSIONAL VELOCITY (MOVING AWAY).

$v \propto d$ (AS $d\uparrow$, $v\uparrow$)

1929: $v = H_0 d$

→ HUBBLE LAW

"HUBBLE FLOW OF GALAXIES"



WHAT DOES MEAN? THE UNIVERSE IS EXPANDING!

1940S: "PLAY THE MOVIE BACKWARDS"

ALL THE DISTANT GALAXIES APPEAR TO BE COMING TOWARDS US → ALL COME TOGETHER AT A POINT IN THE DISTANT PAST.

→ THE UNIVERSE HAD A MOMENT OF CREATION! (NOT ∞ OLD). EVIDENCE? NO BLACK DWARFS.

H_0: HUBBLE CONSTANT. $H_0 \approx 71 \frac{KM}{s \cdot Mpc}$

MOST SOUGHT-AFTER NUMBER IN ASTRONOMY!

TODAY: SOME ASTRONOMERS CALL H_0 THE "HUBBLE PARAMETER" (IT MAY CHANGE).

ON THE "BIG SCALES" (A FEW MILLION ly) GALAXIES TEND TO RESIDE IN GROUPS (GALAXY CLUSTERS). GRAVITATIONALLY-BOUND SYSTEMS OF GALAXIES, OVERRIDE THE EXPANSION OF THE UNIVERSE.

POOR → FEW: LOCAL GROUP (50)
RICH → LOTS: VIRGO CLUSTER (2,500)

LOCAL GROUP:

VIRGO CLUSTER:

IN RICH CLUSTERS (MOSTLY), GALAXY COLLISIONS OCCUR:

FAST: ←●—●→

SLOW: →●←● MERGE

← GALACTIC CANNIBALISM!

SPIRALS: STAR FORMATION (BLUE) } M87: MANY TIMES

- IF WE LOOK AT THE MOTIONS OF GALAXIES WITHIN CLUSTERS, OBSERVED MOTIONS DO NOT MATCH THE COMPUTER MODELS (BASED UPON THE OBSERVED MASS OF THE GALAXIES). GALAXIES APPEAR TO BE "UNDERWEIGHT". IF WE ADD AN ADDITIONAL 90% MASS TO EACH, THE MODELS MATCH THE MOTIONS
  → DARK MATTER!
- FOR SPIRALS WITHIN CLUSTERS: IF WE OBSERVE THE ORBITAL SPEEDS OF THEIR DISK STARS, THEY INCREASE WITH DISTANCE FROM THE NUCLEUS
  → DARK MATTER!

2 POSSIBILITIES: MACHOS, WIMPS ← MOST LIKELY

ONE OF THE TWO BIGGEST AREAS OF RESEARCH IN ASTRONOMY, OTHER IS "DARK ENERGY."

PRIOR TO 2000: THE UNIVERSE IS EXPANDING, SO WILL IT EVER STOP? IS GRAVITY POWERFUL ENOUGH TO HALT THE EXPANSION IN THE FUTURE?

EARLY 2000S: DARK ENERGY (PERMEATES ALL OF SPACE). ACTS LIKE AN "ANTIGRAVITY FORCE." THE EXPANSION IS ACCELERATING!

($E = mc^2$)

UNIVERSE IS:
73% D.E.? 23% D.M.?
4% NORMAL MATTER

IN GALAXY CLUSTERS, WE OFTEN OBSERVE VERY HOT (X RAYS) INTERGALACTIC GAS?

WHY HOT?

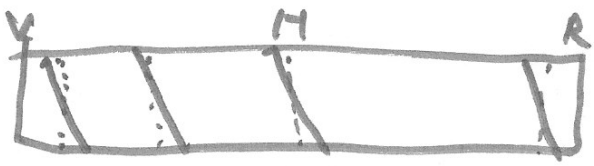
WHY IS IT "METAL RICH"? ← STARS

GALAXIES

TULLEY - FISHER RELATIONSHIP:

GET DISTANCES TO SPIRAL GALAXIES

SPECTRUM OF GALAXY:



THEORY: GREATER THE SLANT OF THE LINES → GREATER THE STARS' ORBITAL SPEEDS → GREATER THE MASS OF THE GALAXY → GREATER ITS TRUE BRIGHTNESS. (THEORETICAL VALUE FOR ITS M)

NOW: MEASURE ITS m (EASY)

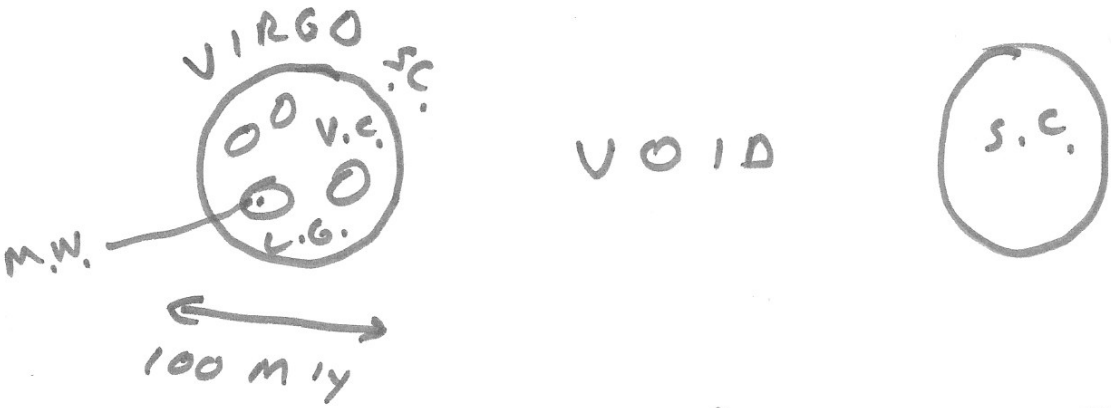
GET DISTANCE FROM:

$$m - M = 5 \log d - 5$$

SUMMARY: MOST LIKELY HAS A BIG ERROR

USEFUL IF THERE ARE NO CEPHEIDS OR TYPE Ia SUPERNOVAE

ON THE "REALLBIG" SCALES (100 MILLION ly),
GALAXY CLUSTERS TEND TO CLUSTER:
→ SUPERCLUSTERS!



ON THE BIGGEST OF SCALES, THE UNIVERSE APPEARS TO BE "BUBBLY" (S.C., VOIDS)

→ THE ORIGINAL BIG BANG THEORY (1940s) DOES NOT PREDICT THIS STRUCTURE.

THIS IS ONE OF THE THREE BIG PROBLEMS WITH THE ORIGINAL BIG BANG THEORY (OTHERS: FLATNESS AND HORIZON PROBLEMS).

- RECENT OBSERVATIONS

① S.C. OFTEN RESIDE IN "FILAMENTS"
→ "COSMIC WEB" (D.E.)

② GREAT ATTRACTOR ?? IT IS AFFECTING 1/3 OF THE VISIBLE UNIVERSE.